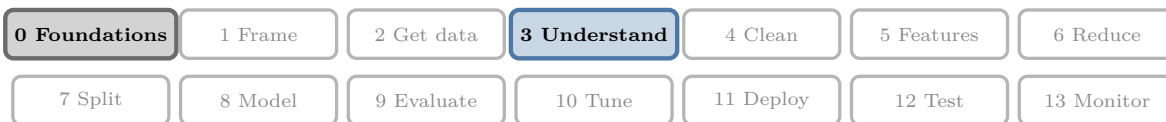


Probability Mass Function and the CDF of a Discrete Variable

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Where this fits: Pipeline map steps 0 (Foundations), 3 (Understand data). See the [Course map](#).



- **Builds on:** Probability density function (PDF) ([Note ML-019](#)); Probability distributions ([Note MA-003](#)).
- **Leads to:** Poisson distribution ([Note MA-022](#)); Standard normal and the z-table ([Note MA-025](#)); P-values ([Note MA-041](#)).
- **Compare with:** Frequency tables ([Note MA-007](#)); Poisson distribution ([Note MA-022](#)); Likelihood ([Note MA-069](#)).

1. Overview

Key point: The PMF gives the probability of each exact value of a discrete random variable (G-617); the CDF adds those probabilities up, giving the probability of a value at most x .

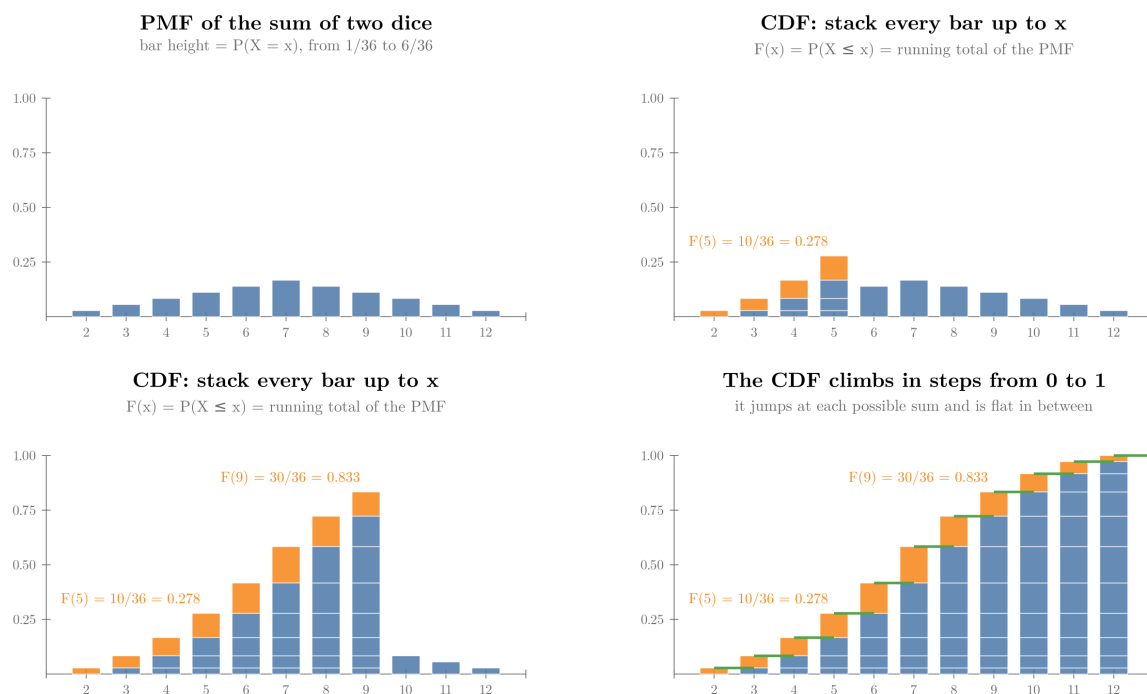


Figure 1: Building the CDF of the sum of two dice by stacking the PMF bars

Figure 1 shows the whole Note in one animation. The blue bars are the PMF of the sum of two dice: the probability of each exact sum. Stacking every bar up to x gives the CDF: the probability of a sum of x or less.

Random variables (G-1620), distributions and the PMF/PDF/CDF family are introduced in the [random variables and distributions Note](#). Here we build the two functions for discrete variables, in this order:

- PMF from a list of outcomes, on three coin flips (section 2);
- PMF from a formula (sections 3, 4 and 6);
- PMF from a simulation (section 5);
- CDF: a running total of the PMF (section 8).

2. From outcomes to a PMF: three coin flips

Key point: List every outcome, group the outcomes by the value of the random variable, and give each value the share of outcomes in its group; the bar chart of those shares is the PMF.

Flip a fair coin three times and let X be the number of heads. X can only be 0, 1, 2 or 3: it cannot be 1.5. How likely is each value? We find out in three steps, shown in Figure 2.

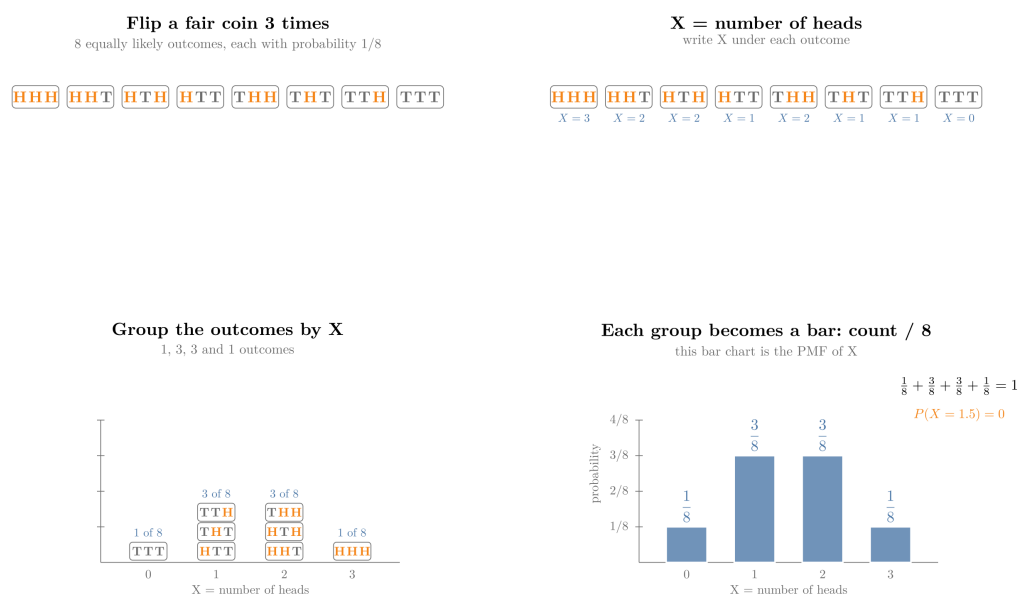


Figure 2: The 8 outcomes of three coin flips are grouped by the number of heads, and each group becomes a bar of height count / 8 (idea after Khan Academy, “Constructing a probability distribution for random variable”)

1. **List every outcome.** Each flip gives H or T, so three flips give $2 \times 2 \times 2 = 8$ outcomes: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT. For a fair coin all 8 are equally likely, each with probability $1/8$.
2. **Write X under each outcome and group them.** One outcome has no heads (TTT), three have one head (HTT, THT, TTH), three have two heads, and one has three heads (HHH).

3. **Give each value its share.** $P(X = 0) = 1/8$, $P(X = 1) = 3/8$, $P(X = 2) = 3/8$, $P(X = 3) = 1/8$.

In the last frame of Figure 2, each group has collapsed into a bar whose height is its share. The bar chart lists every possible value of X and its probability, and there is no bar at 1.5, because $P(X = 1.5) = 0$. A list of probabilities like this, one for each value of a **discrete random variable** (G-617), is called a **probability mass function** (G-1572). The four bars add up to $1/8 + 3/8 + 3/8 + 1/8 = 1$, because one of the four values is certain to happen. Section 3 states the definition in general, and the rest of the Note builds PMFs for dice.

3. The probability mass function

Key point: A PMF assigns a probability to each possible value of a discrete random variable; every probability is at least 0, and together they add up to 1.

The **probability mass function (PMF)** (G-1572) is the probability distribution function of a **discrete** random variable. The PMF assigns a probability to each possible value:

$$p(x) = P(X = x)$$

The formula reads: “the probability that the random variable X takes the value x ”. For the three coin flips of section 2, $p(2) = P(X = 2) = 3/8$. On a graph, the possible values go on the x axis and their probabilities on the y axis.

3.1 The two conditions

Key point: No probability is negative, and the probabilities of all possible values sum to exactly 1.

Not every function can be a PMF. Its probabilities must satisfy two conditions:

1. **In words:** no probability is negative, and the probabilities of all possible values add up to 1, because one of them is certain to happen.
2. **Formula:**

$$p(x) \geq 0 \text{ for every } x, \quad \sum_x p(x) = 1$$

3. **Example:** for one die, every $p(x) = 1/6 \geq 0$, and

$$\sum_{x=1}^6 p(x) = 6 \times \frac{1}{6} = 1$$

For the sum of two dice (the table in the [random variables and distributions Note](#), section 3.1):

$$\frac{1 + 2 + 3 + 4 + 5 + 6 + 5 + 4 + 3 + 2 + 1}{36} = \frac{36}{36} = 1$$

Figure 3 stacks these probabilities into one column for each experiment. Watch the top: six slices of $1/6$, or eleven slices from $1/36$ to $6/36$, both fill exactly the height 1.

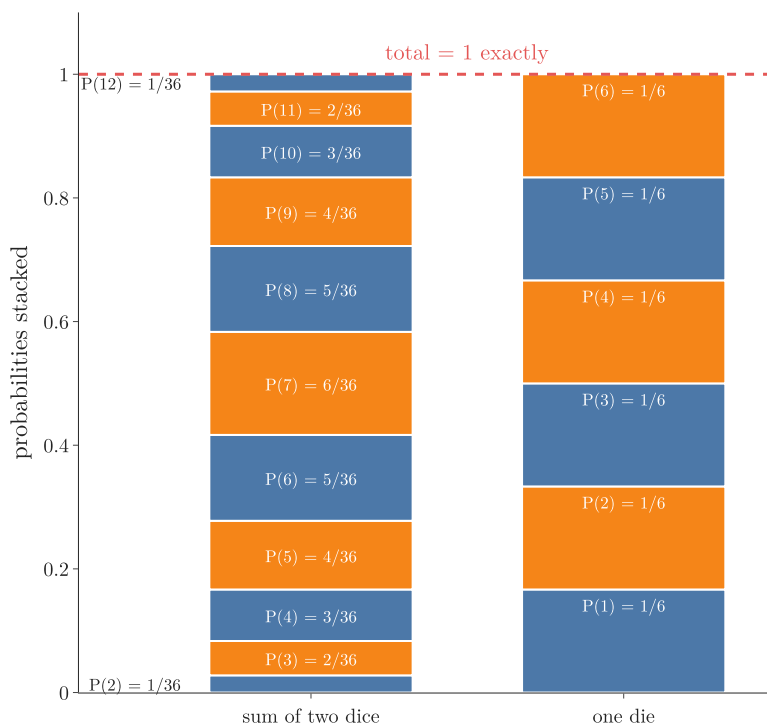


Figure 3: The probabilities of one die and of the sum of two dice, stacked: each PMF fills exactly 1

4. The PMF of one die

Key point: For a fair die the PMF is $1/6$ at each face from 1 to 6 and 0 everywhere else.

A PMF written as a formula $y = f(x)$ must give a value for **every** x , including values the die cannot show. So it has two parts. One new sign appears: $\in$ reads “is one of” or “belongs to”. For example, $3 \in \{1, 2, 3, 4, 5, 6\}$ is true (3 is a face of the die), while $7 \in \{1, 2, 3, 4, 5, 6\}$ is false. The curly brackets $\{ \}$ list the members of a set.

$$f(x) = \begin{cases} \frac{1}{6} & \text{if } x \in \{1, 2, 3, 4, 5, 6\} \\ 0 & \text{otherwise} \end{cases}$$

The formula says at once that every face is equally likely, and that 1.5 or 7 have probability 0. Its graph is six equal bars; a distribution in which every value is equally likely is called a **discrete uniform distribution** (G-618). Figure 4 draws the formula along the whole x axis: six spikes of height $1/6$, and 0 everywhere else, including at 1.5 and 7.

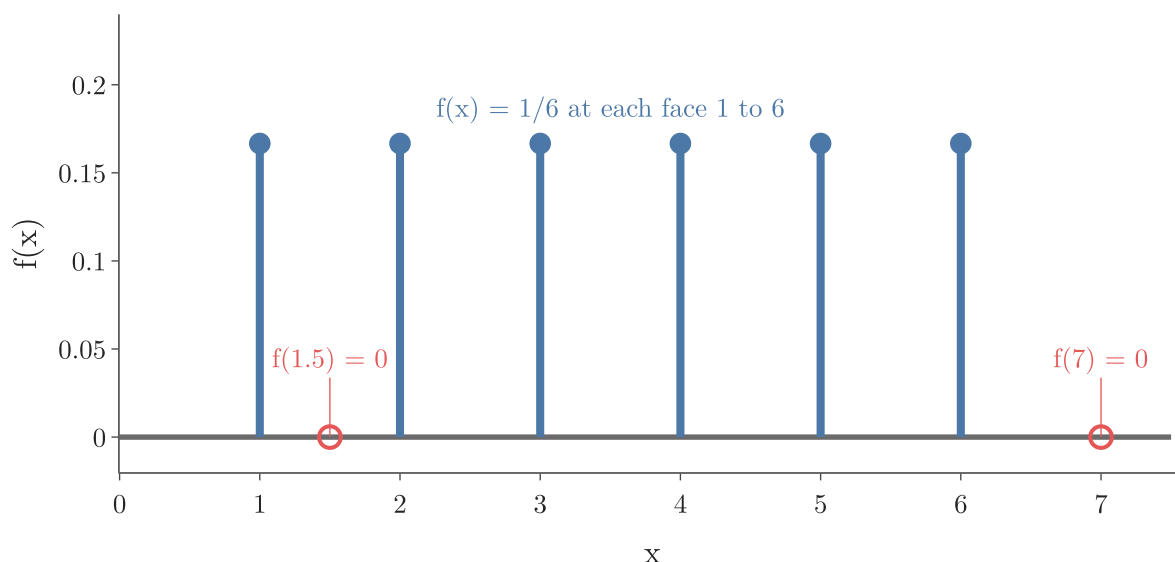


Figure 4: The PMF of one die as a function of every x : $1/6$ at the six faces, 0 elsewhere

5. Estimating a PMF by simulation

Key point: Repeat the experiment many times, count each outcome and divide by the number of trials; the shares approach the true PMF.

We can also find a PMF without any formula: run the experiment many times and count. The share of trials that gave each value is an estimate of its probability, the **estimated PMF** (G-708), the same idea as the relative frequency in the [frequency tables Note](#).

1. **In words:** the estimated probability of a value is the number of times it appeared divided by the number of trials.
2. **Formula:**

$$\hat{p}(x) = \frac{\text{count of } x}{\text{number of trials}}$$

3. **Example:** in 10,000 simulated rolls of one die, the face 1 appeared 1,681 times:

$$\hat{p}(1) = \frac{1681}{10000} = 0.1681$$

The exact value is $1/6 \approx 0.1667$.

The hat on $\hat{p}$ marks an estimate. Figure 5 compares the simulated PMFs (bars) with the exact ones (diamonds), for one die and for the sum of two dice.

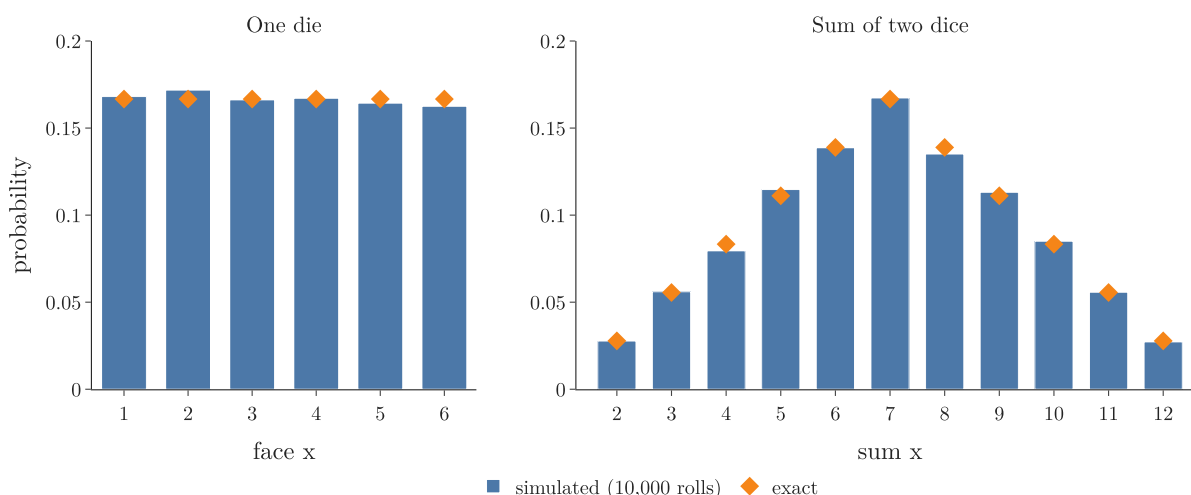


Figure 5: PMFs estimated from 10,000 simulated rolls (bars) and the exact PMFs (diamonds)

The estimates are close but not exact: the six faces came out between 0.1625 and 0.1718, around $1/6$. A random experiment rarely matches theory exactly. For the sum of two dice, the simulated bars follow the same rise to 7 and fall after it as the exact values.

Python: Simulating a die and estimating its PMF.

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(42)          # fixed seed
rolls = rng.integers(1, 7, size=10_000)
# count each face, divide by the total
pmf = pd.Series(rolls).value_counts(normalize=True)
pmf = pmf.sort_index()
pmf.plot(kind="bar")
```

`integers(1, 7)` excludes the upper end, so it returns 1 to 6. `normalize=True` divides the counts by their total; without it we would divide by 10,000 ourselves. `sort_index()` puts face 1 first. For two dice, add two such arrays. The Notebook (`notebook.ipynb`) runs both experiments.

Extra: The more trials, the closer the estimate gets to the true probability. The pull of the estimate towards the true probability is the **law of large numbers** (G-1052; Pishro-Nik §7.1.1). With 100 rolls a face typically comes out anywhere from about 0.12 to 0.22; with our 10,000 rolls all six land within about 0.005 of $1/6$ (the Notebook repeats the 100-roll experiment 1,000 times).

Figure 6 watches the same 10,000 rolls arrive. Watch the title: after 10 rolls face 1 is 0.133 away from $1/6$; after all 10,000, no face is more than about 0.005 away.

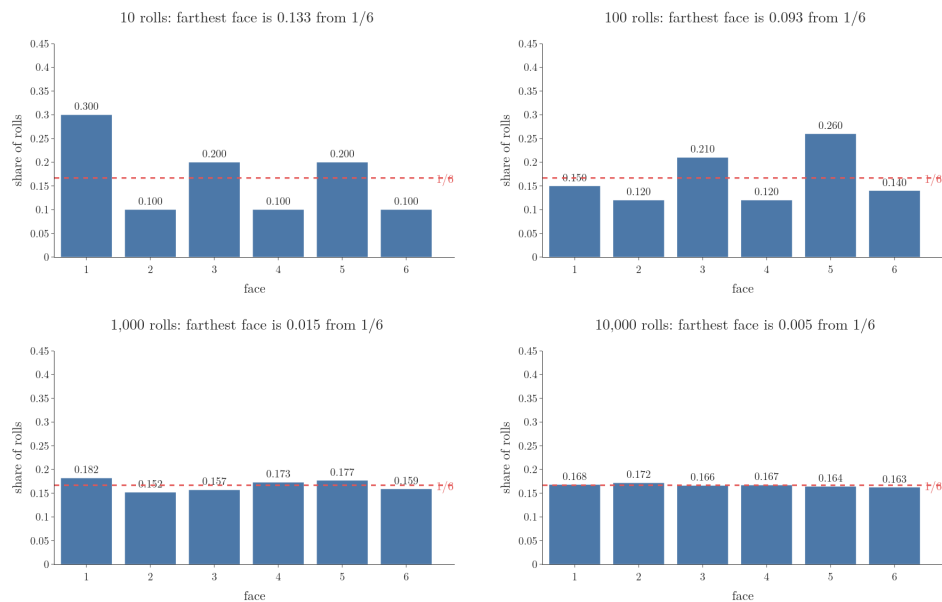


Figure 6: The estimated PMF of one die after 10, 20, 50, $\dots$, 10,000 of the same simulated rolls, against the exact $1/6$

6. The PMF of the sum of two dice

Key point: When the probabilities differ, the PMF formula must give each value its own probability; for two dice one line does it.

The probabilities of the sum of two dice, from $1/36$ for sums 2 and 12 up to $6/36$ for 7, are counted in the [random variables Note](#) (section 3.1) and plotted in Figure 5 (right). Written as a PMF, the list becomes a formula.

Extra: The number of pairs falls by one for each step away from 7, so:

1. **In words:** start from 6 pairs at the sum 7 and subtract the distance from 7.
2. **Formula:** for $x = 2, \dots, 12$,

$$f(x) = \frac{6 - |x - 7|}{36}$$

3. **Example:** for $x = 9$, the distance from 7 is 2:

$$f(9) = \frac{6 - 2}{36} = \frac{4}{36} \approx 0.111$$

Figure 7 shows the rule as a shape: the pair counts form a tent, peaking at 6 for the sum 7 and losing one pair per step away from it.

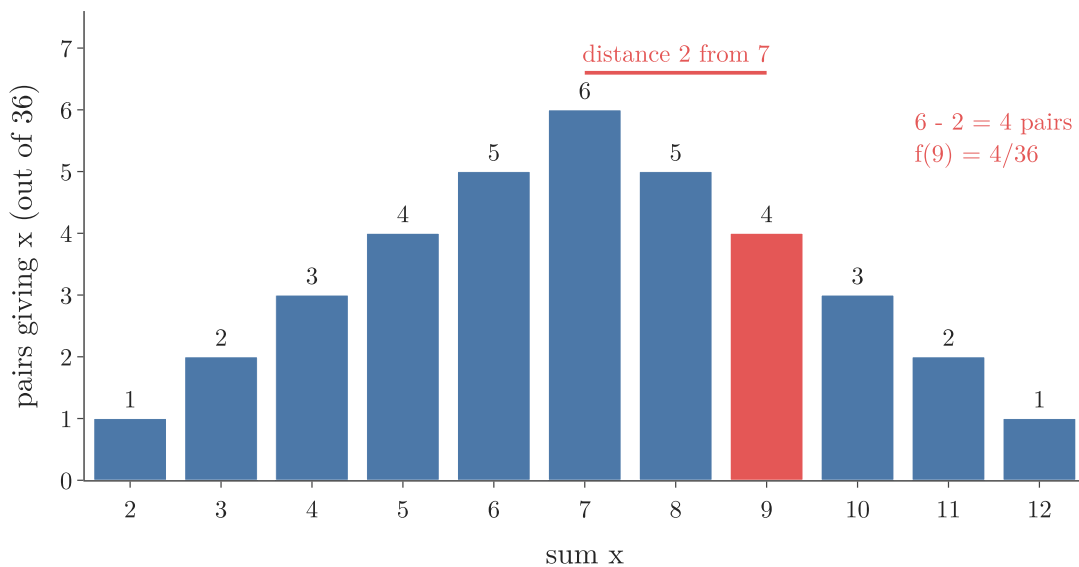


Figure 7: Pairs of dice giving each sum: 6 minus the distance from 7, such as 4 pairs for the sum 9

7. Two famous discrete distributions

Key point: The Bernoulli PMF describes one yes/no trial; the binomial PMF counts the successes in n such trials.

Many discrete experiments follow a famous PMF (see Figure 6 of the [random variables and distributions Note](#)). Two of them come up constantly, Bernoulli and binomial, taught in full in the [Bernoulli and binomial Note](#).

7.1 Bernoulli distribution

Key point: One trial with two outcomes: 1 with probability p , 0 with probability $1 - p$.

The **Bernoulli distribution** (G-275) describes a single trial with two outcomes, success ($k = 1$) and failure ($k = 0$), such as one coin toss.

1. **In words:** success has probability p ; failure has the rest, $q = 1 - p$.
2. **Formula:**

$$P(X = k) = \begin{cases} p & \text{if } k = 1 \\ q = 1 - p & \text{if } k = 0 \end{cases}$$

3. **Example:** with $p = 0.3$, $P(X = 1) = 0.3$ and $P(X = 0) = 0.7$; they add up to 1.

The Bernoulli distribution has **one** parameter, p (with $0 \leq p \leq 1$). The q often listed beside it is not a second parameter: it is fixed by p as $1 - p$. (MML Example 6.8)

7.2 Binomial distribution

Key point: The number of successes in n independent Bernoulli trials with the same p .

The **binomial distribution** (G-308), introduced in the [voting ensemble Note](#), counts the successes in n independent trials, each with success probability p . Its parameters are n and p .

1. **In words:** choose which k of the n trials succeed, then multiply the probabilities of k successes and $n - k$ failures.
2. **Formula:**

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where $\binom{n}{k}$, read as n choose k , is the number of ways to choose k trials out of n .

3. **Example:** the probability of exactly 2 heads in 4 fair coin tosses ($n = 4$, $p = 0.5$). There are $\binom{4}{2} = 6$ ways to place the 2 heads:

$$P(X = 2) = 6 \times 0.5^2 \times 0.5^2 = 6 \times 0.0625 = 0.375$$

Figure 8 draws both PMFs: Bernoulli has just two bars, and the binomial for 4 tosses has five, with 2 heads the most likely at 0.375.

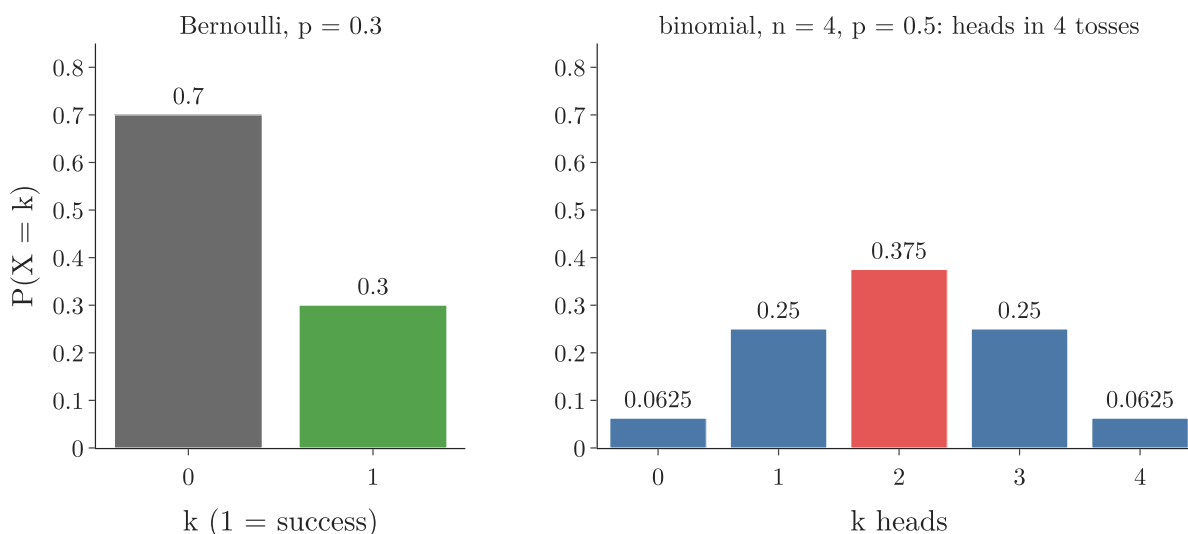


Figure 8: The Bernoulli PMF with $p = 0.3$ (left) and the binomial PMF of heads in 4 fair tosses (right)

8. The cumulative distribution function of a discrete variable

Key point: The CDF $F(x)$ is the probability that the variable is at most x : the running total of the PMF up to x .

The **cumulative distribution function (CDF)** (G-515) of a random variable X gives the probability that X takes a value **less than or equal to** x :

$$F(x) = P(X \leq x)$$

The difference from the PMF is one symbol:

- **PMF at 4:** $P(X = 4)$, the probability of **exactly** 4.
- **CDF at 4:** $P(X \leq 4)$, the probability of 4 **or less**: 1, 2, 3 or 4.

For a discrete variable, “4 or less” means adding the PMF of every value up to 4. That running total is why the function is called cumulative.

1. **In words:** add the probabilities of all possible values up to and including x .
2. **Formula:**

$$F(x) = \sum_{t \leq x} p(t)$$

3. **Example:** for one die,

$$F(4) = p(1) + p(2) + p(3) + p(4) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{4}{6} \approx 0.667$$

The CDF of the die climbs $1/6, 2/6, 3/6, 4/6, 5/6, 6/6$: it starts above 0 and ends at exactly 1.

The discrete CDF is the cumulative relative frequency of the [frequency tables Note](#), with probabilities in place of relative frequencies.

8.1 Reading the CDF of two dice

Key point: The CDF answers “what is the chance of x or less?” in one reading: a sum of 9 or less has probability $30/36$, about 0.83.

Adding up the two-dice PMF gives its CDF:

Sum x	$P(X = x)$	$F(x) = P(X \leq x)$
2	$1/36$	$1/36 = 0.028$
3	$2/36$	$3/36 = 0.083$
4	$3/36$	$6/36 = 0.167$
5	$4/36$	$10/36 = 0.278$
6	$5/36$	$15/36 = 0.417$
7	$6/36$	$21/36 = 0.583$
8	$5/36$	$26/36 = 0.722$
9	$4/36$	$30/36 = 0.833$
10	$3/36$	$33/36 = 0.917$
11	$2/36$	$35/36 = 0.972$
12	$1/36$	$36/36 = 1$

Sum x	$P(X = x)$	$F(x) = P(X \leq x)$
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Figure 9 draws both functions on the same x axis. On the PMF we read the chance of exactly x ; on the CDF, the chance of x or less:

- $F(9) = 30/36 \approx 0.83$: a sum of 9 or less happens in about 83% of rolls.
- $F(5) = 10/36 \approx 0.28$: a sum of 5 or less, in about 28%.

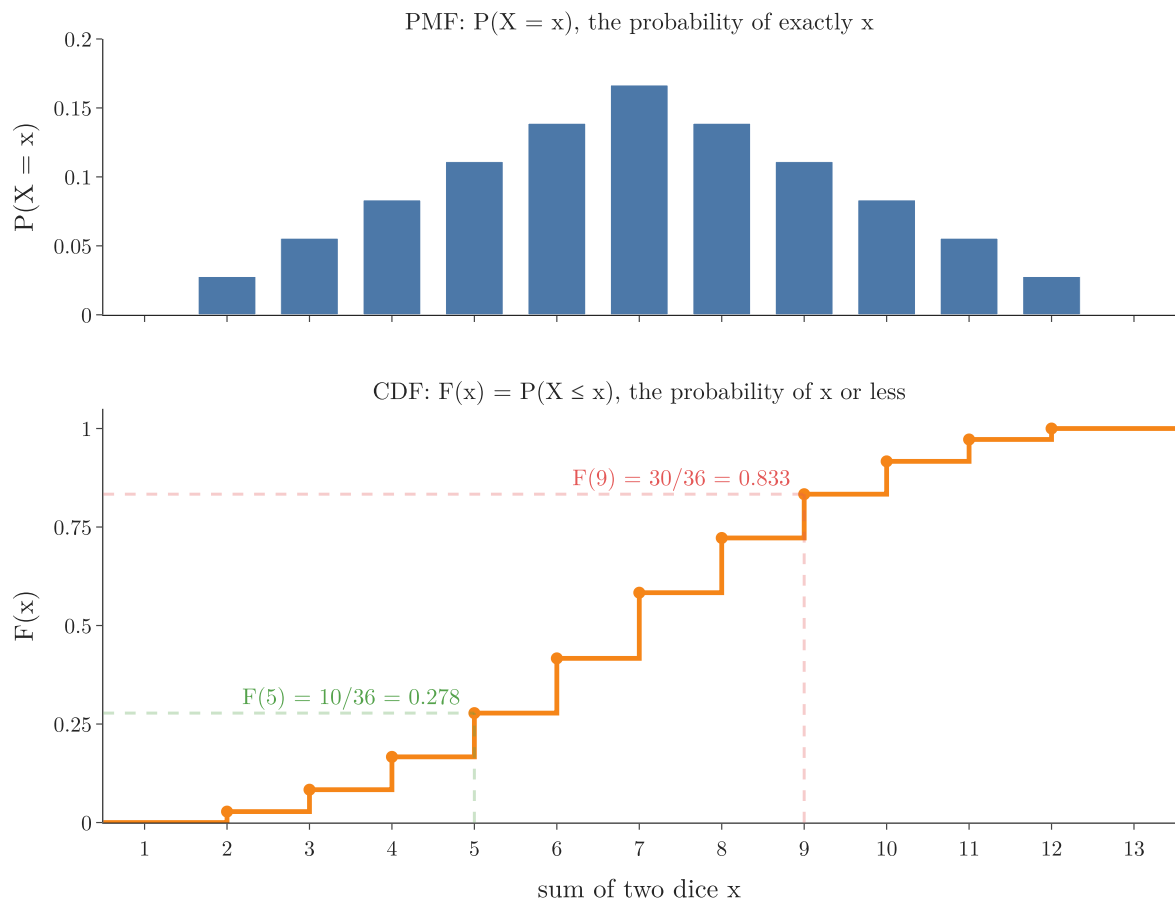


Figure 9: PMF (top) and CDF (bottom) of the sum of two dice

Python: The CDF is the cumulative sum of the PMF.

```
cdf = pmf.cumsum()           # pandas
cdf = np.cumsum(pmf)         # NumPy, same values
```

The simulated CDF from the 10,000 rolls gives $F(9) = 0.832$, against the exact 0.833.

There is no separate name such as “cumulative mass function”: the CDF is called the same for discrete and continuous variables.

Extra: Three facts about every CDF, and two uses of them:

- **A CDF runs from 0 to 1 and never goes down.** Each step adds a probability, which is never negative.
- **A CDF is defined for every x , not only the possible values.** $F(7.5) = P(X \leq 7.5) = P(X \leq 7) = 21/36$, because no sum lies between 7 and 7.5. So the CDF of a discrete variable is a **step function** (G-1889): flat between possible values, jumping up at each one by that value's probability (Figure 9, bottom). Drawing it as bars, one per value, shows only its values at the jumps.
- **The PMF can be read back from the jumps:** the jump at 9 is $F(9) - F(8) = 30/36 - 26/36 = 4/36 = p(9)$.
- **Probability of a range:** $P(a < X \leq b) = F(b) - F(a)$. For a sum above 5 and at most 9:

$$P(5 < X \leq 9) = \frac{30}{36} - \frac{10}{36} = \frac{20}{36} \approx 0.556$$

- **Probability of more than x :** $P(X > x) = 1 - F(x)$. A sum above 9 has probability $1 - 30/36 = 6/36$.

9. Summary

Function	Question it answers	One die	Two dice
PMF $p(x) = P(X = x)$	chance of exactly x	$p(4) = 1/6$	$p(9) = 4/36$
CDF $F(x) = P(X \leq x)$	chance of x or less	$F(4) = 4/6$	$F(9) = 30/36$
Estimated PMF $\hat{p}(x)$	share of trials giving x	$\hat{p}(1) = 0.1681$	$\hat{p}(7) = 0.1673$

- A PMF needs $p(x) \geq 0$ and $\sum p(x) = 1$.
- A PMF formula covers every x : it is 0 for impossible values.
- Simulating many trials and dividing counts by the number of trials estimates the PMF.
- Bernoulli: one trial, one parameter p . Binomial: successes in n trials, parameters n and p .
- The CDF is the running total of the PMF: a step function from 0 to 1.

10. Sources

Built from

- CampusX, “Session 40 - Probability Distribution Functions - PDF, PMF & CDF | DSMP 2023”, YouTube, https://www.youtube.com/watch?v=C_QAURbgBqY
- Khan Academy, “Constructing a probability distribution for random variable”, YouTube, <https://www.youtube.com/watch?v=cqK3uRoPtk0>

Other references

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Example 6.8 (Bernoulli distribution).
- Pishro-Nik, H. (2014). *Introduction to Probability, Statistics, and Random Processes*. Kappa Research. Section 7.1.1 (law of large numbers).

11. Key terms

Term	Meaning
$P(X = x)$	The probability that the random variable X takes the value x
Discrete uniform distribution	A discrete distribution in which every possible value is equally likely, such as a fair die
Estimated PMF	Each value’s share of many repeated trials, used as its probability
Law of large numbers	The more trials, the closer a share of trials gets to the true probability

Term	Meaning
Bernoulli distribution	One trial with two outcomes: 1 with probability p , 0 with probability $1 - p$
$\binom{n}{k}$ (n choose k)	The number of ways to choose k items out of n
Step function	A function that is flat between points and jumps at them, like the CDF of a discrete variable